

Fig. 4: Proposed LED matrix arrangement (3D visualisation in Proteus 8)

To turn on an LED, the corresponding analogue pin is turned to low state and the corresponding digital pin to high state, which turns on the transistor. Thus 5V supply from Arduino reaches the positive terminal of the LED, and current gets a complete closed path via Arduino's analogue pin.

A user-defined function is written in the code so that when we input the value of the row and column

number to the function, corresponding LED in the matrix glows.

J1-J4 are common jumper wires. Interconnect each other through external wires.

Software

Software code for Arduino Uno is written in Arduino programming language and compiled using Arduino IDE. The file name is `led_matrix_7x5_efy.ino`, compiled in Arduino IDE. The program is included in this month's EFY DVD and the algorithm is described in the program using comments. The program is uploaded to Arduino Uno by clicking Upload in Arduino IDE. Select the correct board and COM port before uploading the code.

Construction and testing

An actual-size, single-side PCB of the 7x5 LED matrix is shown in Fig. 2 and its component layout in Fig. 3. Proposed LED matrix arrangement is

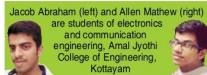
EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

shown in Fig. 4.

Mount the components as shown in the PCB layout. Then, install Arduino IDE in your computer. Open the program and upload it to Arduino board. The LEDs should start glowing.

To test the LED matrix board, check and ensure each LED is in its respective row and column. For initial testing of Arduino board, upload a simple program like Blink from the example sketches in Arduino IDE. **EFY**



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